Supplementary material: pre-analysis plans, addenda and replication notes

The files below are reproduced verbatim; SHA-256 hashes are listed in Section 6. Code and data will be deposited in a public repository with a DOI on acceptance; the code is available to the editor and referees on request before then.

1. Replication notes

The replication package, to be deposited on acceptance, is organised as follows.

Run everything from this folder's root.

- `pap/` pre-analysis plans (01 Tel Aviv; 02 Gulf, hashed before any Gulf gap statistic) and addenda B and C (exploratory), with SHA-256 hashes in `HASHES.txt`.
- `code/00–08*.py` Tel Aviv pipeline (`python code/run_all.py`), from the official TASE records in `data/raw/tase/`.
- `code/gcc/09–17*.py` Gulf pipeline and exploratory checks; `code/pooled/20_figures_tables.py` figures and tables.
- `output/` results files and LaTeX tables from which every number in the paper is taken.

Data not included: - Gulf share prices (Tadawul, DFM, ADX) are licensed from LSEG Workspace and may not be redistributed. `lseg_jobs/` would retrieve them with a Workspace licence; `code/gcc/10_build_gcc.py` expects them in `data/raw/lseg/`. The cell-level statistics derived from them are in `output/`. - NYSE session dates (S&P 500 index, `data/raw/tase/GSPC.csv`) and the Yahoo Finance TA-125 series used in the appendix on missing sessions are from Yahoo Finance and are not redistributed; download ^GSPC and ^TA125.TA daily history.

2. Pre-analysis plan 01 (Tel Aviv)

Pre-analysis plan

Working title: Time or trading? Evidence on weekend return variance from the Tel Aviv Stock Exchange's move to a Monday–Friday week.

Written: 2026-09-23, before any estimation on outcome variables. Descriptive work done so far: tabulating trading days per weekday, and checking official TASE prices against Yahoo Finance on three dates. No variances, regressions or mean returns have been computed.

1. Setting

Until 1 January 2026 the Tel Aviv Stock Exchange (TASE) traded Sunday to Thursday. Since 5 January 2026 it trades Monday to Friday. The Friday session closes with an auction at 13:45 Israel time, before the NYSE opens. Monday–Thursday hours did not change: the closing auction ends at about 17:25, roughly one hour after the NYSE open. Sunday sessions closed at about 15:50 from 8 March 2020.

Consider each non-trading interval ("gap"), from the TASE close on the previous session to the TASE open on the current session.

- The **weekend gap before the switch** (Thursday close to Sunday open) lasts about 64.6 hours. It contains about 1.86 NYSE regular sessions: the Thursday session from about 10:25 New York time, plus all of Friday.
- The **weekend gap after the switch** (Friday close to Monday open) lasts about 68.2 hours. It contains exactly one NYSE session, Friday.
- **Ordinary weeknight gaps** last about 16.6 hours and contain about 0.86 NYSE sessions.

- The **Sunday-night gap before the switch** (Sunday close to Monday open) lasts about 18.2 hours and contains no NYSE session.

The switch therefore holds the calendar length of the weekend roughly fixed (+6%) while nearly halving the foreign trading time inside it. The pre-switch Sunday night provides a gap of weeknight length with no foreign trading.

2. Hypotheses

Let $g_t = \log(O_t/C_{t-1})$ be the gap return and $d_t = \log(C_t/O_t)$ the intraday return of the TA-125 index.

- **H-cal (calendar time).** Gap variance is proportional to elapsed calendar hours H_t.
- **H-info (foreign information arrival).** Gap variance is proportional to the foreign trading time U_t inside the gap, measured in NYSE sessions.

Predictions for the relative weekend variance, $R = \text{Var}(g \mid \text{weekend})/\text{Var}(g \mid \text{ordinary weeknight})$, before and after the switch: - under H-cal, R is about 3.9 before and about 4.1 after; - under H-info, R is about 2.2 before and about 1.2 after.

3. Data

- Official TASE end-of-day data (base, open, close, high, low) for TA-125 (primary) and TA-35 (robustness), 2005-01-02 to 2026-09-23, from the TASE website's historical-data service.
- S&P 500 from Yahoo Finance (^GSPC). Used only to date NYSE sessions and to compute NYSE close-to-close returns. The Yahoo record for TASE is **not** used for estimation, because it omits Sunday sessions (see §6).
- NYSE session times are 9:30–16:00 New York time. Half-days are counted as full sessions (a known small error). Clock times use zoneinfo (Asia/Jerusalem, America/New_York), so the offsets are correct when the two countries' daylight-saving changes fall on different dates.

4. Sample

- **Primary sample:** 2020-03-08 to 2026-09-23, the period with constant Sunday hours. Pre = before 2026-01-05; post = from 2026-01-05.
- **Gap types:**
  - W-pre: first session after a Friday–Saturday closure, Thursday close to Sunday open.
  - W-post: Friday close to Monday open.
  - N: ordinary weeknight, next calendar day, the previous session not a Sunday before the switch, and not a Friday after it.
  - S-pre: Sunday-night gap, Sunday close to Monday open.
  - H: any other gap, including holidays. H gaps are excluded from the main tests and kept for the structural model.
- **Robustness samples:**
  - (a) excluding 2020;
  - (b) excluding 2023-10-07 to 2023-12-31 (outbreak of war);
  - (c) excluding June 2025 (Israel–Iran war);
  - (d) the long pre sample, 2005-01-02 to 2025-12-31, with approximate hours.

5. Specifications (primary first)

P1. Variance ratio test. Normalise by the local volatility of the index: $z_t = g_t/\hat{\sigma}_t$, where $\hat{\sigma}_t^2$ is the mean of d_s^2 over the 22 sessions before t. Estimate R_{pre} and R_{post} as ratios of mean z^2 for W versus N gaps. Test H_0: $R_{post} = R_{pre}$, the change predicted to be zero under H-cal. - Inference: bootstrap that resamples weeks (stationary block bootstrap, mean block length 4 weeks, 9,999

draws), plus a regression-based test. - Regression: $z_t^2 = a + b\,W_t + c\,\text{Post}_t + \delta\,W_t \times \text{Post}_t + e\,S_t + u_t$ on N, W and S-pre gaps, with HAC standard errors (Newey–West, 10 lags). - The test is on δ. H-info predicts $\delta < 0$; the magnitude implied by the prediction ratios is roughly a halving of R. - Report the result in levels as well, together with a Brown–Forsythe test.

P2. Structural decomposition. Pool all gaps of type N, W and S, together with post Thursday nights (a type N gap). Estimate

$$E[z_t^2] = \kappa + \lambda_H H_t + \lambda_U U_t$$

by OLS with HAC standard errors. Identification of λ_H separately from λ_U comes from the S-pre gaps and the switch. - Report the implied share of weekend gap variance attributable to calendar hours. - Pre-registered expectation: $\lambda_U > 0$ and λ_H small. Any outcome is reported.

P3. Loading on US returns. Regress $g_t = \alpha + \sum_k \beta_k X_{k,t} + \varepsilon_t$ on the NYSE close-to-close returns of the sessions inside the gap: for W-pre the US Thursday and Friday returns, for W-post the US Friday return. The timing prediction is that the pre-switch weekend gap loads on US Thursday and Friday, and the post-switch gap only on US Friday. - Then compare the variance of the residuals of W versus N gaps with the P1 procedure. This asks whether any excess weekend variance remains after projecting out US information.

P4. Means (secondary, known low power). Mean of g_t and of the close-to-close return for the first session after the weekend, pre versus post, reported with standard errors. No claim will rest on this test.

P5. Data-vendor exhibit (descriptive). Compare Yahoo Finance ^TA125.TA with the official series: - count the sessions that Yahoo omits; - run a standard day-of-week regression on each source, 2014–2025, and report the difference.

6. Robustness (fixed list)

1. TA-35 in place of TA-125.
2. Outcome g_t^2 without normalisation.
3. $\widehat{\sigma}_t$ computed from 66 sessions instead of 22.
4. Samples (a)–(d) above.
5. Winsorising z_t at the 1st and 99th percentiles.
6. Excluding gaps adjacent to Israeli holidays.
7. Post-period placebo: the switch date moved to 2025-01-05 on pre data only.

7. Multiple testing and reporting

- The headline test is P1 (δ); everything else is secondary.
- Every result above will be reported whether or not it is significant. Any additional analysis will be labelled exploratory.
- A null result on P1, given the power of about 37 post-switch weekends, would mean the post-switch sample cannot yet separate the hypotheses. It will be reported with a minimum detectable effect.

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Addendum A (2026-09-23; written after the index-level P1–P5 results, before any stock-level data were analysed)

Why. The index-level P1 test has low power: 34 weekends after the switch give a bootstrap SE of about 0.45 on the log-ratio. The index-level results also showed that the Sunday-night gap before the switch, which contains no NYSE cash session, had variance similar to ordinary weeknights. This raised

A1. Stock-level triple difference (the new primary stock-level test).

- Universe: current TA-125 constituents, with daily TASE data from 2020-01-01. Survivorship is noted as a limitation.
- Groups:
  - DL: TASE's own "Dual-Listed Company" flag, 16 firms. These shares trade abroad, mostly on NASDAQ/NYSE, dur-

A2. Alternative foreign-trading clock (exploratory). U^{fut} = hours of CME equity-index futures trading inside the gap. Re-estimate P2 with U^{fut} in place of U.

A3. Robust scale (exploratory). Rerun P1 with $\log|z|$ as the outcome, as a heavy-tail-robust scale comparison.

3. Pre-analysis plan 02 (Gulf)

Pre-analysis plan 02 — the Gulf trading-week switches (UAE 2022, Saudi Arabia 2013)

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Written 27 September 2026, before any gap-return variance for a Gulf market has been computed. At the time of writing the only Gulf data examined are the outputs of LSEG job 01 (session dates by weekday and the share of index opens equal to the previous close), which

Plan 01 (`01_pap_TASE.md`, with Addendum A) governs the Tel Aviv switch and is not changed by this document, except that §6 below adds the pooled test.

1. Question

Whether the variance of returns over a market closure accrues in calendar time or in trading time abroad. A trading-week switch keeps the weekend closure at roughly the same length but changes how many US cash sessions it contains; the old or new calendar also contains weeknight closures with no US session.

2. Events and regimes

market	last day old week	first day new week	old week	new week
UAE: DFM and ADX	Thu 30 Dec 2021	Mon 3 Jan 2022	Sun–Thu	Mon–Fri
Saudi Arabia: Tadawul	Wed 26 Jun 2013	Sun 30 Jun 2013	Sat–Wed	Sun–Thu

Session hours and dates are taken from exchange notices and verified against the session dates present in the data (step 1). If a verified date differs from this table, the verified date is used and the difference is reported; nothing else in the plan changes.

Windows. - UAE: primary window 2 Jan 2018 – 25 Sep 2026 (four years before, all available after). Robustness: drop 23 Feb – 30 Jun 2020 (Covid crash); symmetric window of 3 years 9 months either side. - Saudi Arabia: primary window ±4 years (1 Jul 2009 – 30 Jun 2017). Robustness: ±2 years (1 Jul 2011 – 30 Jun 2015), which ends before the opening to qualified foreign investors in June 2015; ±5 years.

3. Units and data

Primary unit: constituent stocks. For each market (DFM, ADX, Tadawul), the 30 largest firms by market capitalisation on the last day of the primary window whose price history covers the whole primary window, taken from LSEG. This list is fixed by rule before any gap statistic is computed and is saved to `data/raw/lseg/`. The survivorship this implies is stated in the paper; it is not expected to bear on a ratio of closure variances.

Secondary unit: the index (DFM General Index, FTSE ADX General Index, TASI). If, in any year of a window, more than 5% of an index's opens equal the previous close to six digits, index results for that market are reported only in an appendix.

Observation. A gap g = log(open_t) − log(close_{t−1}) on consecutive sessions of the same security. Excluded: a gap where either price is missing, where volume on day t is zero, or where the open equals the previous close exactly and volume on day t is below the stock's 5th percentile (a no-trade open).

Local scale. $\hat{\sigma}^2$_t is the mean of squared open-to-close log returns over the 22 sessions before t (at least 15 required). z^2_t = g^2_t / $\hat{\sigma}^2$_t, winsorised at the 99.9th percentile of the market's pooled z^2.

4. Closure types

Each closure (close of t−1 to open of t) is described by - H: calendar hours; - U: the number of NYSE cash sessions inside it, counting a session fraction by the share of its hours that fall inside the closure, NYSE half days counted at their length; - W: 1 if the closure contains at least one day that is a weekend day of the market's calendar in force.

Types: **N** ordinary weeknight (W = 0, U = 1); **Z** weeknight with no US session (W = 0, U = 0: the UAE Sunday night before the switch; the Saudi Saturday and Sunday nights before, the Sunday night after); **W** weekend closure with no holiday; **H** any closure containing an exchange holiday (excluded from the primary tests).

Implied values (to be recomputed from verified hours):

	UAE before	UAE after	Saudi before	Saudi after
weekend U	2 (Thu, Fri)	1 (Fri)	3 (Wed, Thu, Fri)	2 (Thu, Fri)
Z closures	Sun night	none	Sat, Sun nights	Sun night

5. Hypotheses and tests (per market)

Let R_k = mean z^2 over type-k closures divided by mean z^2 over type-N closures, within market × regime.

H1 (weekend). Under trading time the weekend ratio falls when the weekend loses a US session: Δ log R_W = log R_W(after) − log R_W(before) < 0. Under calendar time Δ log R_W = log(H_W(after)/H_W(before)), close to zero. One-sided test at 5%.

H2 (zero-US weeknight). Under trading time R_Z < 1; under calendar time R_Z ≈ H_Z / H_N ≈ 1. One-sided test of log R_Z < 0 in every regime that has Z closures.

Inference. Stock level: z^2 regressed on type × regime dummies with standard errors clustered two ways (calendar week, firm); ratios and their log-differences by the delta method. Index level and as a check: stationary block bootstrap over calendar weeks, mean block length 4 weeks, 1,999 draws. Firm-level randomisation inference is not available here (no dual-listed/domestic split), so none is claimed.

Secondary (reported, not tested for significance). The same ratios computed on log|z| (median-based), which is robust to the heavy tails of gap returns; and close-to-close weekday variances.

6. Pooled test across the three switches

The model E[z^2 | closure] = a_m + b·U + c·W, with a market-specific intercept a_m and common b and c, is estimated on the TASE, UAE and Saudi stock panels pooled (TASE: domestic and dual-listed separately, since the TASE plan already treats them separately). Primary pooled tests: 1. b > 0 (one-sided); 2. the calendar-time restriction that every ratio equals its hours ratio, R_k = H_k/H_N for all k, as a joint Wald test (two-sided).

The switch-by-switch estimates of Δ log R_W are combined by inverse-variance weighting and shown together in one figure. No switch is dropped from the pooled analysis because of its result.

7. Exploratory (labelled as such in the paper)

- A second clock: ICE Brent futures trading hours inside each closure in place of U.
- Heterogeneity by firm size, foreign ownership limit or sector (banks, real estate, petrochemicals).
- The Friday session in the UAE after 2022, compared with Monday–Thursday.

8. What would count against the claim

Δ log R_W ≥ 0 in both Gulf switches, or R_Z not below 1 where Z closures exist, or b not positive in the pool. Any of these would be reported in the abstract.

4. Addendum B

Addendum B to plan 02 — exploratory analyses decided after unsealing

Written 27 September 2026, after the primary results of plan 02 were computed (output/results_gcc.json). Everything below is exploratory and will be labelled so in the paper. It was prompted by an adversarial review of those results.

Primary results as unsealed (stock level, 22-day denominators, winsorised): UAE ΔlogR_W = −0.07 (0.11); KSA −0.05 (0.22); combined with TASE −0.07 (0.08). R_Z: UAE pre (Sunday nights) 1.18; UAE post (US-holiday nights) 0.70; KSA pre 0.72, post 0.74; TASE domestic 1.08, dual-listed 0.53. H1 not supported in any market; H2 supported in KSA (both regimes), UAE post, TASE dual-listed; contradicted in UAE pre.

B1 Bound. Per market, the weeknight per-session effect δ = 1 − R_Z and the weekend per-session effect implied by the switch, δ_W = R_W,pre (1 − e^Δ)/ΔU, with delta-method SEs; a 95% upper bound on δ_W from the combined estimate. B2 US holidays. Drop 24 Dec–2 Jan and Good Friday; report Monday US holidays alone. B3 UAE Sunday nights. (i) Sunday vs other weekdays: volume, share of zero intraday returns, intraday variance; (ii) reversal: gap on the preceding session's open-to-close return, by night; (iii) denominator built without Sunday sessions; (iv) Sunday-night z^2 by tercile of Sunday volume relative to the firm's median. B4 Denominators: 66-session window. B5 Saudi price limits: drop gaps with |g| ≥ 9.5%. B6 Ramadan: drop Ramadan months. B7 TASE

confound: the post-switch Israeli weekend closure contains Sunday, a working day; stated in the text, no test.

5. Addendum C

Addendum C — further robustness and scale checks (exploratory)

Decided after the primary results of plan 02 and Addendum B were known. Exploratory throughout. Code: code/gcc/13_review_checks.py, 14_review_round2.py, 15_abs_scale.py, 16_median_bootstrap.py, 17_close_to_close.py.

C1 Saudi post-period split at the QFI opening (15 June 2015). C2 Exclusion of the weeks of 7 Oct–31 Dec 2023, 13–30 Apr 2024 and 13–30 Jun 2025. C3 Scales that down-weight the tail: (E|z|_k/E|z|_N)^2 on all gaps; exp(2(E log|z|_k − E log|z|*N)) on non-zero gaps; median-based ratios (plan 02 secondary) with a cluster bootstrap over calendar weeks (999 draws); heterogeneity (Q, I^2, random effects). C4 First night of the week: ordinary weeknights by the weekday of the following session. C5 Share of the sum of z^2 contributed by the largest 1% of share-gaps; frequency of gaps above the pre-switch 99th percentile. C6 Close-to-close weekend measure: log(C_t/C*{t−1}) normalised by the mean squared close-to-close return of the 22 previous sessions, on the mean and |c| scales, to count information priced during the first session after a closure.

6. SHA-256 hashes

```
b089bc82e7dba744996645fd57659fa8306bdd982f36f233a484878bdc8e100a  02_pap_GCC.md
Sat Sep 26 20:52:25 UTC 2026
d5c787829c6df2826517d091d4cb5832f6e30bb8619aa80475f49fdf46a6d202 03_addendum_B_exploratory.md
b62149dc2acf183528d7c35e34813ecb2c008c36a4419cf3e4d7dac9290e5359  04_addendum_C.md
```